

RESEARCH ARTICLE



Invasive South African populations of the freshwater snail *Tarebia granifera* belong to a single globally distributed lineage within a species complex

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Abstract

The Asian thiarid snail *Tarebia granifera* is a highly successful invasive species and has had substantial impacts in South African freshwater ecosystems, often out competing and replacing native snail populations. However, there is limited understanding of the molecular diversity of the snail beyond its native Southeast Asian range, with few attempts to identify origins or potential number of invasion events in to new freshwater environments. Similarly, identifying *T. granifera* via morphological methods is complex and a method by which several efforts to monitor the spread of the snail and the parasites they transmit have depended upon. By employing mitochondrial cytochrome c oxidase subunit I DNA barcoding, phylogenetic and species delimitation analyses it has been possible to identify the occurrence of three putative species within *T. granifera*. However, crucially, the phylogenetic and population genetic analyses revealed only a single lineage to contain all invasive haplotypes of *T. granifera* including those from South Africa. Also, the population genetic analyses revealed a single haplotype in the South African populations, suggesting a rapid expansion event into several aquatic ecosystems across the country. The current findings are consistent with a single invasion event; however, the current study highlights the inability of single marker techniques to reveal the fine-scale evolutionary history of invasion for *T. granifera*. Developing genomic tools is crucial to monitor the spread of snail species across southern Africa, to assess their ecological impact and for trematode disease surveillance.

Keywords *Tarebia granifera* · Invasive · Lineages · Cryptic species · South Africa · DNA barcoding

Introduction

Invasive species are second only to habitat destruction in their impact on native biodiversity (Dueñas et al. 2021; Weyl et al. 2020). Within freshwater ecosystems, invasive species can replace native species, as well as becoming ecosystem

engineers, leading to habitat degradation resulting in alteration of the biomass of consumers and even affecting nutrient fluxes within an environment (Gerber et al. 2025; Moslemi et al. 2012). However, biological invasions by cryptic species further complicate the issue, as morphologically similar but genetically distinct taxa can differ considerably in their ecological impact, invasion dynamics and responses to management, which could lead to an underestimation of biodiversity loss and misattribution of invasion effects (Bickford et al. 2007; Pfenninger and Schwenk 2007). Because morphologically similar lineages within aquatic mollusc species complexes can be indistinguishable, invasive lineages and multiple, independent invasions of such lineages can escape early detection and monitoring, enabling them to spread extensively before recognition and complicating risk assessments and biosecurity efforts (Carlton 2009; Geller et al. 2010). This can be further complicated when invasive species also have the capacity to alter disease transmission within a habitat,

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either by acting as a novel compatible host for native pathogens or by introducing novel pathogens into an environment (Thieltges et al. 2025). This is particularly true for invasive freshwater molluscs, with high reproductive rates, tolerance to a wide range of environmental conditions and being a crucial part of trematode life cycles, making invasions by such organisms a global One Health concern.

The parthenogenic snail *Tarebia granifera* (Lamarck, 1816) represents one of the most successful invasive molluscs of tropical and subtropical freshwater systems and has been suggested to be a complex of multiple cryptic species (Pearson et al. 2024; Veeravechsukij et al. 2018). The snail has a natural range from India through to the mainland of Southeast Asia, with northern range in China and Taiwan, through to the Philippines and south to the Indonesian Archipelago and the islands of the Western Pacific (Veeravechsukij et al. 2018). Within its native range, *T. granifera* functions as an important first intermediate host for zoonotic trematodes within the Heterophyidae and Opisthorchiidae, including the cancer-causing liver fluke *Clonorchis sinensis* (Paller et al. 2021; Pearson et al. 2024). As a consequence of the international aquarium trade, *T. granifera* is now found in several states in the USA, many of the Caribbean islands, Central and South America, Israel and sub-Saharan Africa (Pearson et al. 2024; Veeravechsukij et al. 2018).

Thiarid species are notoriously challenging to identify accurately and *T. granifera* is no exception owing to extensive phenotypic plasticity, particularly in shell morphology. Veeravechsukij et al. (2018) showed that across the natural range of *T. granifera* in Thailand and Malaysia there was significant morphological variability within populations, with several characteristics overlapping with other thiarid snail species. However, snails morphologically identified as *T. granifera* based on shell structure, brood pouch morphology and internal anatomy were genotyped using the mitochondrial cytochrome c oxidase subunit I (COX1) DNA barcoding region and the resultant phylogenetic reconstruction showed at least two distinct lineages. The analyses revealed that there was at least a 13.8% divergence between the lineages, which also appeared to separate geographically based on snail populations from the north and the south that represented not only different altitudes but also different habitats in Thailand. This led to Veeravechsukij et al. (2018) concluding that *T. granifera* should be considered a cryptic species made up of a complex of at least two genetically distinct species but also highlighted that deeper analyses were required to confirm this to be the case. This difficulty in defining *T. granifera* increases the challenge of attempting to not only identify the snail to species level but also identifying invasion routes and the potential origins of establishing populations. As freshwater snails are also known to invade localities through multiple independent invasion events, the occurrence of cryptic invasions, where another genetically distinct lineage of a species invades a

location where the same non-native species exists, frequently go undetected, as has been described in other molluscs of medical and veterinary importance, including Physidae snails invading Africa (Lawton et al. 2018; Albrecht et al. 2025), apple snail *Pomacea* sp. invading South America and Asia (Lv et al. 2013) and *Melanooides tuberculata* into the Africa and the USA (Tolley-Jordan et al. 2023).

Although Veeravechsukij et al. (2018) highlighted the existence of at least two distinct mitochondrial lineages of *T. granifera*, there have been no studies that have attempted to identify which of these occurs outside of its native range beyond Southeast Asia nor have any studies applied molecular techniques to identify the occurrence of multiple cryptic invasion events. South Africa has seen a major invasion event of *T. granifera*, which have established in numerous water bodies across South Africa since it was originally identified in KwaZulu-Natal in 1999 (Appleton and Nadasan 2002), with a particularly high prevalence in the northeast of the country (Malherbe 2018). Furthermore, recent surveys in Mozambique and Zimbabwe have confirmed the presence of *T. granifera* at multiple freshwater sites within these countries (Appleton et al. 2009; Gerber et al. 2024). The invasion success of *T. granifera* within South African freshwater ecosystems provides the ideal scenario to test the hypotheses that (i) invasive populations of *T. granifera* outside its native range, including South Africa, are derived from a single lineage of the species complex previously identified by Veeravechsukij et al. (2018) and that (ii) multiple genotypes from a single lineage of the *T. granifera* complex have invaded South Africa, with establishment through more than one founder event. By undertaking comparisons of publicly available *T. granifera* and COX1 sequences generated here, this current study employed DNA barcoding and phylogenetic approaches to accurately identify the number of cryptic lineages within the snail populations collected from freshwater ecosystems of South Africa.

Material and methods

Sampling and snail collections

Tarebia granifera were collected from sites in the northern and eastern parts of South Africa (Fig. 1). These include seven sites on the Limpopo River (LIMP-2) system, including five tributaries: the Mogalakwena (MOGA), Luvuvhu (LUVU), Olifants (OLIF-1 and OLIF-2), Groot-Letaba (G-LETA) and Letaba (LETA) rivers (all in the Limpopo province), as well as from a single site (PHON-4) on the Phongolo River (KwaZulu-Natal province) during surveys in 2021 and a single site in the North West province during 2024 (Fig. 1). Snails were hand-picked from vegetation and stones for 30 min, as described by Nguyen et al. (2021).

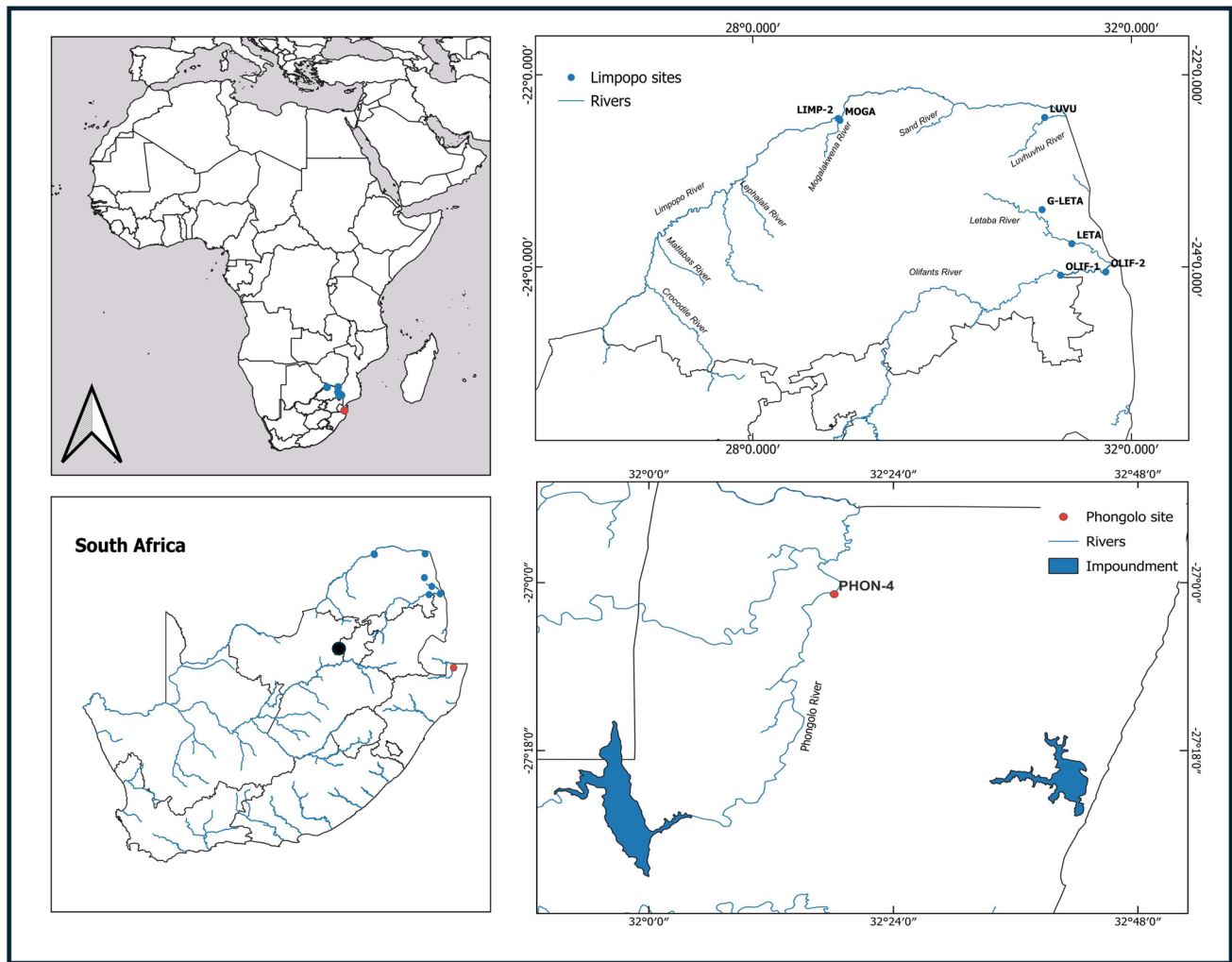


Fig. 1 Map indicating the sampled sites in South Africa. Sites in the Limpopo River system are highlighted with blue points and those in the Phongolo River system with red points. The North West province sample is indicated by the black point on the map

Larger individuals (> 10 mm – mature adults) were targeted, since they have a higher likelihood of trematode infection (Giannelli et al. 2016). Individual snails were then fixed in labelled tubes with 96% ethanol and preserved in a mobile freezer (−4 °C) on-site until genetic analyses at North-West University (NWU), Potchefstroom. All samples were collected and processed with the necessary permits (permit no. ZA/LP/107276) from the Limpopo Department of Economic Development, Environment and Tourism and Ezemvelo KZN Wildlife (permit no. OP 864/2021), as well as ethical clearance from the NWU AnimCare Animal Research Ethics Committee (ethics no. NWU-01539-20-A9).

DNA extraction, PCR amplification and sequencing

A total of 35 individual snails were sequenced, representing the seven sampling sites sampled from the Limpopo River system and from the Phongolo River and five individuals

from North West province. Snail shells were carefully crushed with pliers and the entire snail removed. A small piece of each snail's foot tissue and hepatopancreas were removed using sterile forceps and scalpel blades (Schwelm et al. 2020; Nguyen et al. 2021). Equipment was sterilised with bleach, then rinsed with sterile distilled water and finally washed with 96% molecular-grade ethanol following each dissection to prevent cross-contamination to ensure polymerase chain reaction (PCR) efficacy (Schols 2019). The tissue samples were preserved in separate tubes with 96% molecular-grade ethanol and stored in a freezer (−20 °C) until further analysis. Hepatopancreas samples were divided in two, with one half crushed and placed in separate sterile 1.5ml microcentrifuge tubes for further analysis.

The ethanol was decanted and the remaining ethanol was evaporated in a heat block at 56 °C. Total genomic DNA was extracted from both snail foot tissue and

hepatopancreas samples using the PCR Biosystems Rapid DNA Extraction Kit (PCR Biosystems, Analytical Solutions, Randburg, South Africa). Following the manufacturer's protocol, 20 µl lysis buffer, 10 µl proteinase K buffer and 70 µl molecular-grade water were added to each sample. Samples were incubated at 75 °C for 5 min (with vortex mixing twice) and then at 95 °C for 10 min. Following incubation, samples were diluted with 150 µl molecular-grade water, instead of the recommended 900 µl, to obtain higher DNA concentration. Samples were centrifuged at 15,596 RCF for 1 min and 150 µl of the supernatant transferred to sterile tubes and frozen (−20 °C) to prevent DNA degradation.

The mitochondrial COX1 is the only marker to date used to study the population genetics and phylogenetics of *Tarebia*, thus the molecular DNA barcoding method used by Lawton et al. (2015, 2018) was employed in this current study. Primers LCO1490 (5′-GGTCAACAAATCATAAAGATATTGG-3′) and HCO2198 (5′-TAAACTTCA GGGTGACCAAAAAATCA-3′) as described by Folmer et al. (1994) were used to perform PCR in 25 µl reactions. An aliquot of 5 µl of the resultant amplicon was visualised on 1% agarose gels stained with GelRed. The remaining 20 µl of PCR positive samples were then sequenced using fluorescent dye terminator sequencing kits and run for analyses on an Applied Biosystems 3730XL automated sequencer. Primer sequences were removed from the 5′ ends of the raw sequences during editing and assembly in Bioedit (Hall 1999), with resultant sequences of up to 655 bp prior to alignment.

Phylogenetic analyses and species delimitation

Initial identification of the *T. granifera* sequences was completed by undertaking a simple BLASTN search performed against GenBank through the National Center for Biotechnology Information site (<http://www.ncbi.nlm.nih.gov/>) confirming all individuals to belong to this species complex. All available COX1 sequences for *T. granifera* were retrieved for comparison, representing snail populations from across their native range in Asia, including Thailand, Malaysia, Singapore, Timor and China, as well as non-native ranges including USA, Tahiti and Moorea (Supplementary Table S1). Using *Thiara amarula* as an outgroup, an alignment containing a total of 139 *T. granifera* COX1 sequences was created using MUSCLE (<http://www.ebi.ac.uk/Tools/msa/muscle/>). Phylogenetic analyses of these sequences were performed to identify the occurrence of distinct snail lineages that could be associated with invasions. Phylogenetic analyses were performed using maximum likelihood (ML) approaches

implemented through MEGA 11 (Tamura et al. 2021). It is also important to note that owing to the fragment analysed being a short portion of COX1, the sequences were analysed using a single unpartitioned dataset under one best-fit substitution model, the standard approach for invertebrate DNA barcoding. The Kimura 2-parameter model was identified as the most appropriate nucleotide substitution model for phylogenetic reconstruction using the Bayesian information criterion (BIC), and 1000 bootstrap replicates were used to calculate nodal support, both of which were implemented in MEGA 11.

Lineage identification and species delimitation

Several complementary approaches were employed to assess species delimitation across the COX1 dataset, owing to the cryptic nature of Thiaridae snails. Although these approaches are not statistically independent, each of the delimitation methods make different assumptions of the alignment, and congruence of the results from each of the methods enables a more robust inference than a single method. Their combined outputs were used to infer the occurrence of distinct molecular operational taxonomic units (MOTU) and ultimately potential candidate species. Initially, simple pairwise *p*-distances were calculated between lineages and clades in MEGA 11 to assess percentage of divergence, using the 3% divergence concept within COX1 DNA barcodes to define putative distinct species, as suggested by Hebert et al. (2003). This heuristic rule of thumb was applied to provide an initial, exploratory indication of divergence between clades. However, more robust evidence was sought through the 4× ratio, as described by Birky et al. (2010). This criterion derives from population-genetic theory and was developed specifically for species detection and identification in asexual organisms; it compares the ratio of the mean pairwise differences between mitochondrial DNA sequences from two distinct clades (*K*) with the mean pairwise differences within a clade (*Θ*), with $K/\Theta > 4$ between clades representing two distinct species. The larger value produced in the final calculation is considered to represent the true difference between species, reducing the likelihood of a false positive. The 4× ratio has previously been applied to thiarid snails of the genus *Aylacostoma*, for which genetically distinct species were also recognised from mitochondrial sequence data (Vogler et al. 2014, 2016). Thirdly, the distance-based automatic barcode gap discovery (ABGD) method was employed on the whole of the *T. granifera* COX1 alignment and implemented through Spart explorer server (spartexplorer.mnhn.fr). The ABGD analyses identified the genetic barcode gap between intra- and inter-specific variation. As the analyses use pairwise genetic distances

between snail COX1 sequences, it does not require prior species assignments; this then allows the sequences to be recursively separated into candidate putative species based on statistically supported thresholds of $P < 0.01$. Finally, species delimitation based on tree topology was performed using the Poisson tree processes (PTP) maximum likelihood model as applied to the ML phylogenetic reconstruction based on COX1 individual haplotypes and implemented through the species delimitation server (species.h-its.org). The PTP analyses infer species boundaries from differences in branching rates across the tree, assuming substitutions accumulate according to a Poisson process but at different rates within and between species, with short branches considered indicative of within species coalescent processes and longer branches indicative of between species diversifications. These multiple approaches for species delimitation were applied owing to the parthenogenic nature of *T. granifera* and the recognised limitations of single locus analyses. All MOTUs recovered in this study were treated as candidate species and considered putative.

Inter-population genetic variation

To identify the occurrence of multiple incursion events into South African environments, standard population genetic analyses were performed to assess the diversity within and between *T. granifera* lineages and countries. Unique COX1 haplotypes were identified from the full alignment in DnaSPv6 (Rozas et al. 2017), and simple population genetic comparisons were performed between COX1 sequences generated from different geographical isolates using standard measures of molecular diversity. It is important to note that when the analyses considered the entire global population of *T. granifera*, sequences from China, USA, Tahiti and Moorea were included; however, they were omitted from any country-level analyses, as each of these countries was represented by a single COX1 sequence. Owing to the invasive nature of *T. granifera*, analyses were performed to identify signatures of demographic changes in the snail's population. Pairwise mismatch analyses were performed to identify the occurrence of population expansion events in DnaSPv6. The distribution of frequency of pairwise differences between sequences within each lineage was performed for under 1000 replicate simulations. Subsequently, the raggedness (r) statistic was used to measure the smoothness of observed mismatches to test for any substantial departure from unimodality. Additionally, both the Fu's F_s and Tajima's D were calculated for each geographical isolate and lineage, to identify any significant departure from neutrality as another indicator of population expansion events. Finally, to provide insights into the structure and association of populations, the

evolutionary inter-relationships between unique *T. granifera* COX1 haplotypes most parsimonious haplotype networks were constructed with TCS as implemented through PopArt (Leigh and Bryant 2015).

Results

Phylogenetic analyses and evidence of cryptic species in *Tarebia granifera*

The 139 COX1 sequences used in this study represented 17 distinct haplotypes, and the ML phylogenetic analyses split *T. granifera* into three distinct MOTUs within two well-supported lineages, with nodal bootstrap values of 100 and both lineages containing snails from across Asia (Fig. 2). Lineage 1 contained snail sequences from Thailand and Malaysia, with lineage 2 being composed of snails from a wider geographical spread, including not only Thailand and Malaysia but also Timor, Singapore, China, Tahiti, Moorea, USA and South Africa. Lineage 2 was further split into two sub-clades, with all snails sampled from invasive populations, including those from South Africa, only being found within sub-clade A (Fig. 2). Both sub-clade A and B had sequences from Malaysian snail populations in basal positions as well as representatives scattered throughout.

To determine the occurrence of putative distinct species, initially genetic p -distance was calculated between each of the MOTUs. When lineage 1 was compared with each of the sub-clades within lineage 2, the p -distances ranged between 13.5% and 13.7% divergence (Fig. 3A). This was contrasted when both sub-clade A and B were compared with only 3.18% divergence, but both comparisons did exceed the Hebert et al. (2003) 3% minimum divergence to indicate distinct species (Fig. 3A). The ABGD analyses also revealed three initial and recursive partitions supported by $P = 0.001$, with each of the partitions corresponding to the three MOTUs identified in the phylogenetic analyses (Fig. 3B). This was further supported by the PTP analyses which also highlighted the MOTUs as distinct. Lineage 1 was strongly supported as a distinct species with ML support estimates of 0.9. Similarly, sub-clade A within lineage 2 was also well supported as a potential distinct species with moderate support at 0.8 (Fig. 3B). However, sub-clade B had comparatively lower support with values of 0.5. Finally, the 4 $\times$ ratio analyses were applied to determine occurrence of speciation across the lineages. The analyses did identify the lineages as representing separate species (Table 1). Interestingly, as with the ABGD and PTP analyses, the 4 $\times$ ratio also separated sub-clade A and B in lineage 2 as distinct MOTUs representing separate species with $K/\theta = 7.624$,

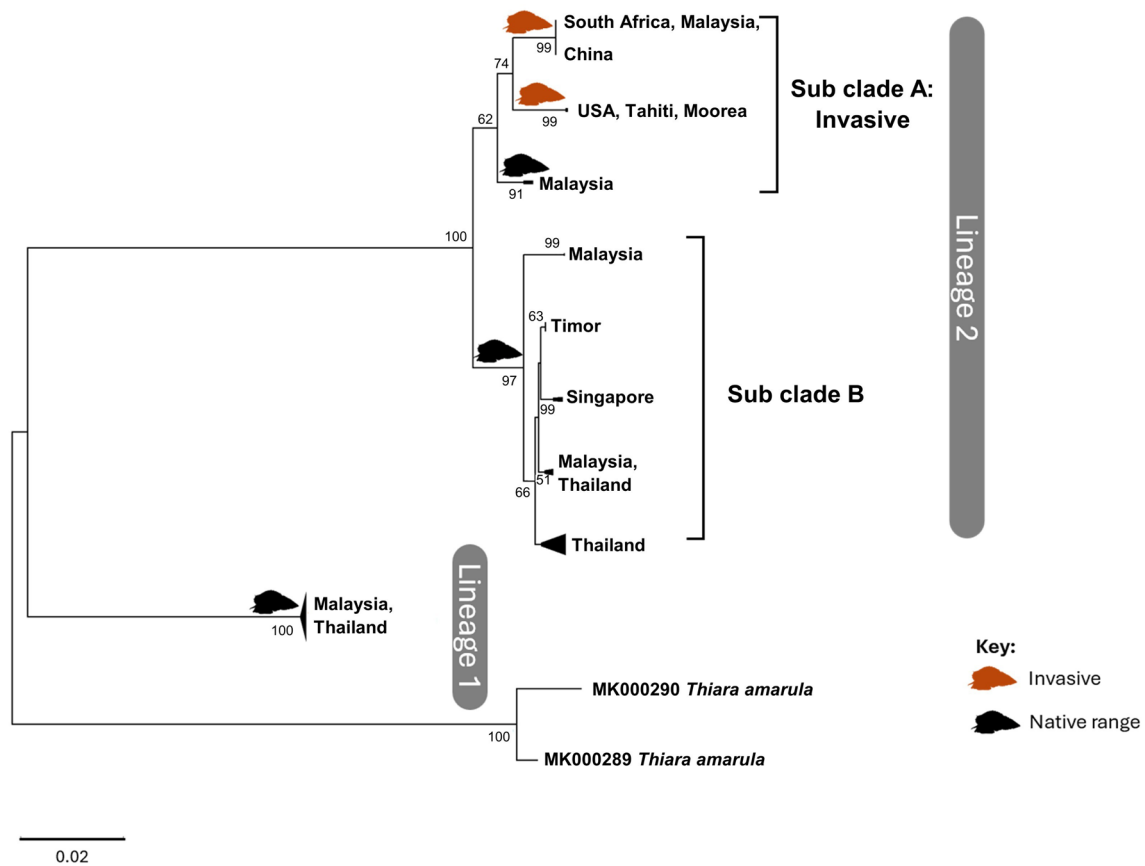


Fig. 2 Maximum likelihood phylogenetic reconstruction of *Tarebia granifera*, where phylogenetic reconstruction was performed using the K2P substitution model and all nodal support values ≥ 50 are shown. Owing to the large number of sequences used, taxa were col-

lapsed where possible, and diversity is represented by shapes from vertical lines indicating low diversity through to triangles indicating the occurrence of diversity

substantially above the minimum of four needed to distinguish between species (Table 1).

Inter-relationship between geographical isolates of *Tarebia granifera*

Population genetic analyses of *T. granifera* showed there to be high levels of variation in the COX1 of individuals from Malaysia and Thailand, with the highest haplotype diversity seen in Malaysia. The snails collected across South Africa appear to be genetically identical with a single haplotype shared across all sites (Table 2). To account for the fact that several of the geographical isolates, especially those from Malaysia and Thailand, would indeed represent a mixture of species, each of the lineages were also analysed as large independent populations. Lineage 1 appeared to have the lowest level of genetic variation with $\pi = 0.0022$. Sub-clade B had the highest nucleotide diversity with $\pi = 0.0059$ and

the highest number of haplotypes as well as the greatest haplotype diversity ($h = 9$, Hd 0.81). Sub-clade A had the lowest number of haplotypes and haplotype diversity ($h = 4$, Hd = 0.24) despite not having the lowest nucleotide diversity at $\pi = 0.00471$ (Table 2).

Haplotype network analyses also confirmed the occurrence of two major lineages, with lineage 2 being separated into two sub-lineages and haplotypes considered to be generated from invasive snail populations only occurring within sub-clade A (Fig. 4). Haplotypes from Thailand and Malaysia were scattered across the haplotype network, occurring in each of the lineages. However, there was an absence of haplotypes from Thailand within the invasive sub-clade A. The analyses also illustrate that the sole haplotype found in South Africa is shared with snail populations from Malaysia and China (Fig. 3A, B) and shared a predicted recent common ancestor with a haplotype found in invasive *T. granifera* populations from the USA, Moorea and Tahiti.

Fig. 3 Genetic divergence and species delimitation of the cryptic species of *Tarebia granifera*, where **A** is a comparison of genetic *p*-distances between lineages, showing the 3% divergence species delimitation limit as described by Hebert et al. (2003), and **B** is an unrooted maximum likelihood phylogenetic reconstruction based on the COX1 haplotypes identified in this study. Each clade identified as a putative species is highlighted by red branches. Grey columns highlight the species partitions identified by the ABGD (light grey) and PTP (dark grey) analyses, with nodal support values representing the PTP, and maximum likelihood support estimates for species delimitation were as follows: ≥ 0.9 is strong support, ≥ 0.8 is good to moderate support and ≥ 0.5 is reasonable support but requires evidence from elsewhere

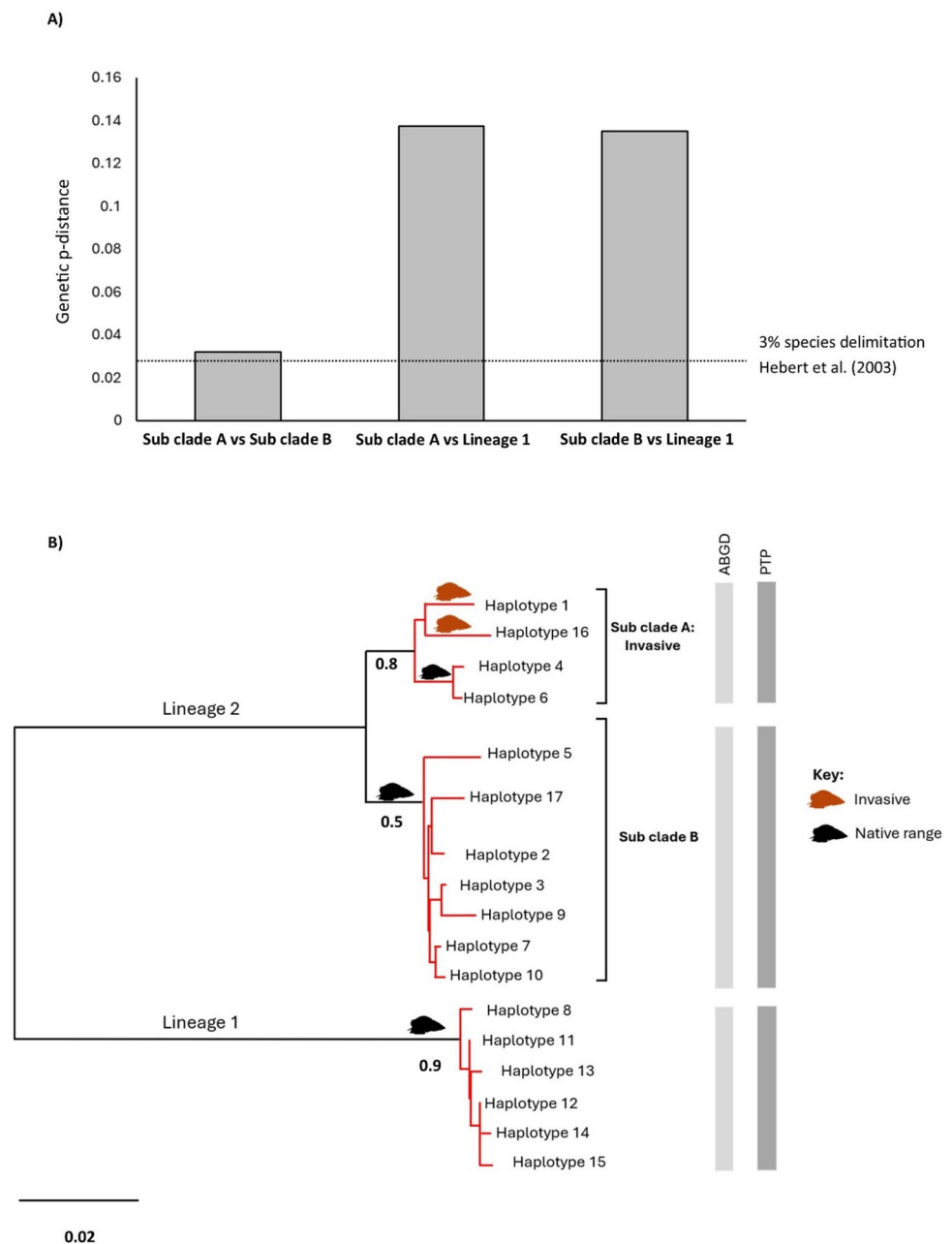


Table 1 The 4× ratio calculation for species delimitation between lineages and sub-clade of *Tarebia granifera*

	Sub-clade A	Sub-clade B	Sub-clade A	Lineage 1	Sub-clade B	Lineage 1
Number of sequences	52	45	52	47	45	47
$K = D =$ total pairwise differences between clades	0.0334		0.1404		0.1315	
$d =$ total pairwise differences within clades	0.00427	0.01101	0.00427	0.00224	0.01101	0.00224
$\pi = (d \times (n/n - 1))$	0.00435	0.01126	0.00435	0.00229	0.01126	0.00229
$\Theta = (\pi / (1 - (4\pi/3)))$	0.00438	0.01143	0.00438	0.0023	0.01143	0.0023
K/Θ	7.624		61.037		57.168	

N numbers of sequences, *K* total pairwise differences between clades and *d* pairwise differences within clades. The highest value of *K*/ Θ represents the most conservative estimation of species differentiation

Table 2 Genetic variation of COX1 in *Tarebia granifera*

Sample	Number of sequences	S	h	Hd	Pi	Fu's <i>F</i>	Tajima's <i>D</i>	Raggedness (<i>r</i>)
All	139	98	17	0.87	0.07	42.17	3.5557*	0.0684
South Africa	35	0	1	0	0	na	na	na
Timor	11	0	1	0	0	na	na	na
Malaysia	18	98	9	0.9	0.05	7.949	0.4186	0.0537
Thailand	67	88	9	0.86	0.06	39.587	3.6516*	0.0673
Singapore	4	4	3	0.83	0.035	0.134	−0.7801	0.3056
Lineage 1	47	5	6	0.78	0.0022	−0.152	0.5221	0.0577
Lineage 2 sub-clade A	47	17	4	0.24	0.00471	4.426	−1.0252	0.6334
Lineage 2 sub-clade B	45	18	9	0.81	0.0059	0.656	−0.6861	0.1598

Populations from China, USA, Tahiti and Moorea are omitted from country-specific analyses, as they are represented by a single COX1 sequence. *S* number of mutations, *h* number of haplotypes, *Hd* haplotype diversity, *Pi* genetic diversity.

*Significant *P*-value ≤ 0.01

Demographic history of *Tarebia granifera*

Owing to the identification of distinct lineages that could represent putative species, genetic demographic analyses were performed at the lineage level. These analyses revealed conflict between the Fu's *F* and Tajima's *D*, with lineage 1 having a negative Fu's *F* but a positive Tajima's *D*, and the reverse was seen in both sub-clades in lineage 2, with no significant scores found. Pairwise mismatch analyses revealed lineage 1 to be unimodal, indicating a high frequency of haplotypes with few mutations (Fig. 5). Both sub-lineages A and B of lineage 2 appeared to be multimodal but still with low frequency of highly divergent haplotypes with high pairwise differences (Fig. 5). The raggedness index (*r*), the comparison of experimental distribution of mismatch pairs, was not significantly different from observed distributions, with each of the lineages indicating the potential of population expansion events (Table 2).

Discussion

The success of an invasive species is ultimately linked to its adaptability, including its tolerance of a wide range of environments, reproductive rate and high levels of phenotypic and genetic diversity, all of which can mediate success. Crucially, the ability to identify distinct cryptic lineages and species is paramount, as genetically distinct populations of exotic and/or pest species may differ substantially in characteristics that enable tolerance of a range of environmental conditions, affect fecundity and ultimately establish new populations in naive ecosystems (Xuan et al. 2025). This is a particular problem for the Thairidae, especially *T. granifera*, which is taxonomically challenging even in its native range

throughout Southeast Asia (Veeravechskij et al. 2018). The phylogenetic and haplotype network analyses presented in this current study confirmed the existence of the two discrete lineages that had already been identified by Veeravechskij et al. (2018) and further resolved lineage 2 into two sub-clades A and B.

All haplotypes included within this study had previously been characterised in Veeravechskij et al. (2018), and the phylogenetic analyses presented in this study are also congruent with that of Veeravechskij et al. (2018), providing reassurance in analytical methods and validity in approach. However, unlike the Veeravechskij et al. (2018) study, the analyses presented here suggests that the two sub-clades of lineage 2 may represent distinct MOTUs and could be considered as putative distinct species within the *T. granifera* species complex. The separation of these two clusters was also documented by Veeravechskij et al. (2018); however, their analyses focused predominantly on the separation between the two major clades and not between the sub-clades within them. In this current study, the authors argue that sub-clade A and B within lineage 2 could be representative of two distinct species owing to the congruence of the species delimitation approaches applied here: inter-clade *p*-distances exceeding the 3% COX1 barcode threshold suggested by Hebert et al. (2003), with 13.5–13.7% divergence between lineage 1 and each sub-clade, and 3.18% between clades A and B; ABGD partitioning into three recursive partitions at $P = 0.001$; and the Birky et al. (2010) 4× ratio, with K/θ values of 7.624, 61.037 and 57.168 for the sub-clade A/B, sub-clade A/lineage 1 and sub-clade B/lineage 1 comparisons, respectively. The PTP analyses likewise recovered three putative species, albeit with weaker support for sub-clade B (0.5) than for lineage 1 (0.9) and sub-clade A (0.8), which is consistent with relatively recent divergence

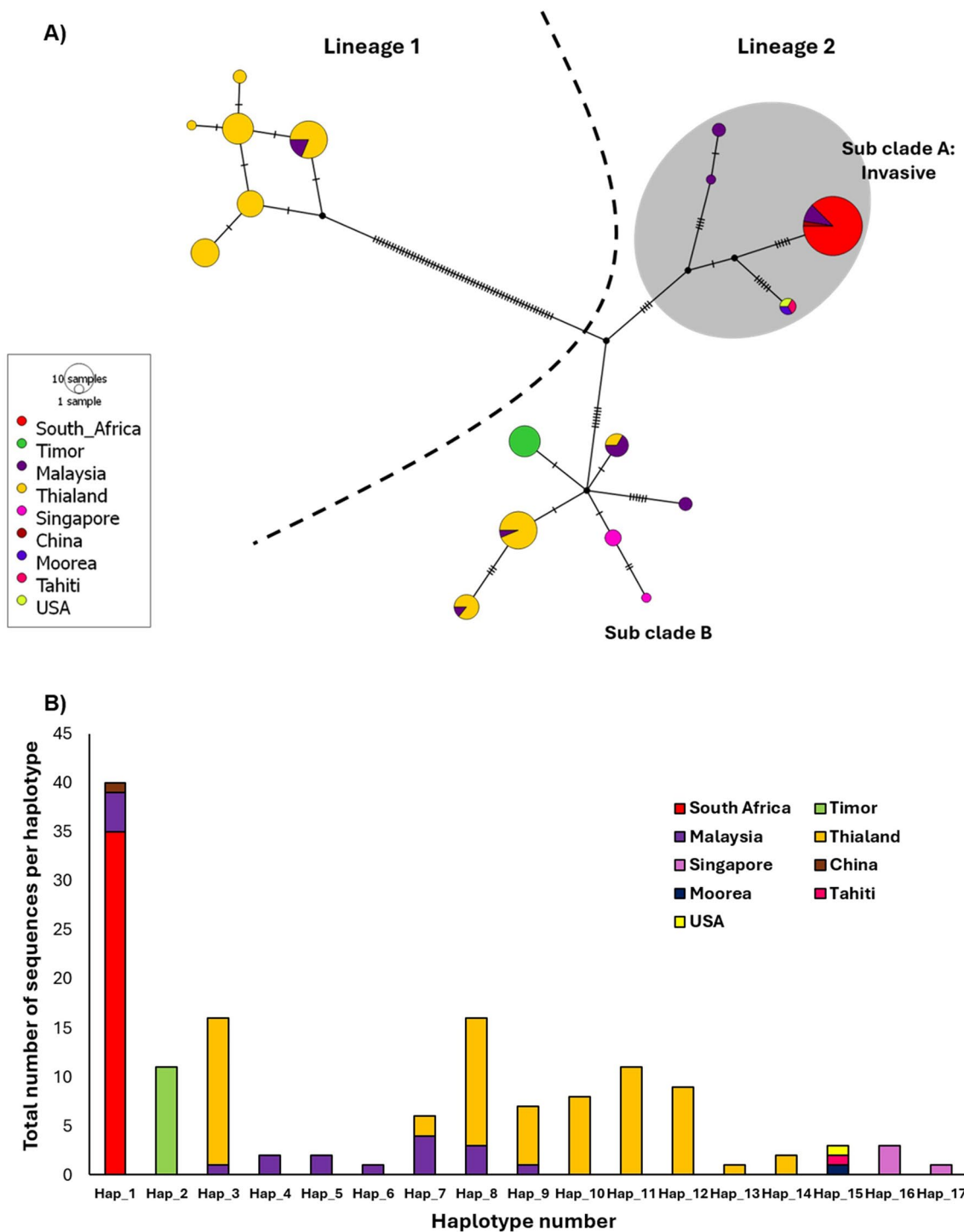


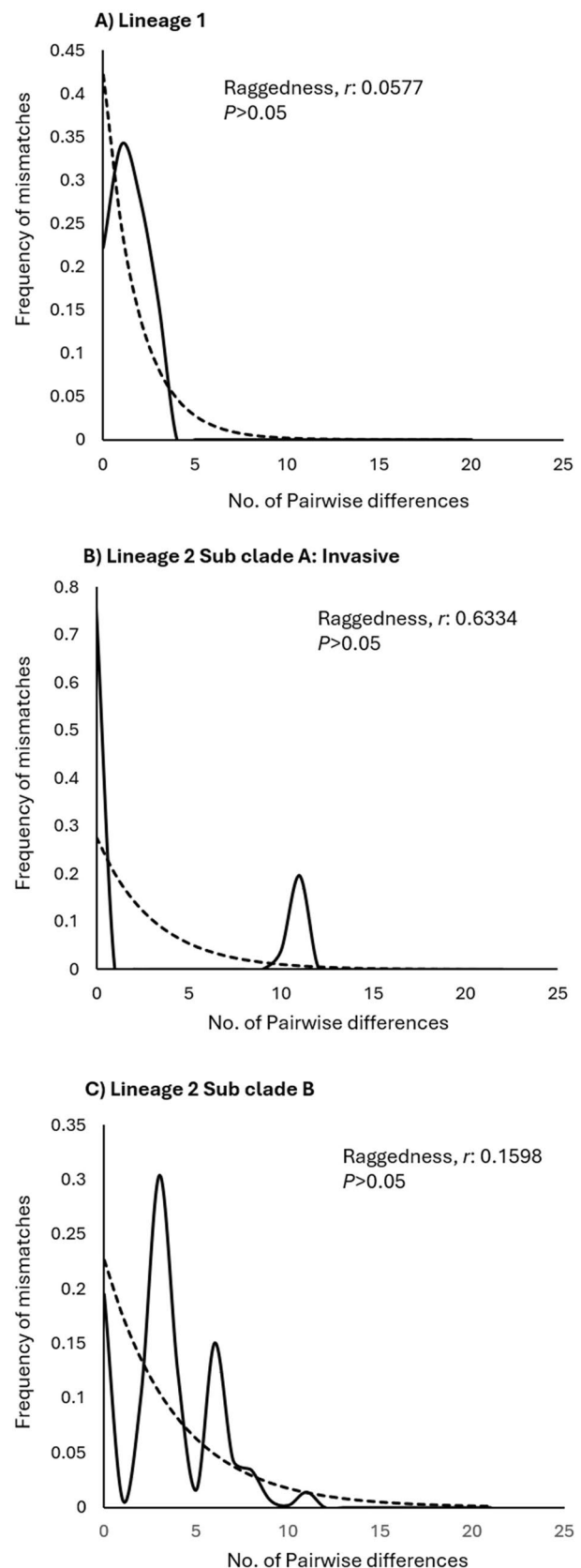
Fig. 4 Diversity, frequency and inter-relationships of *Tarebia granifera* COX1 haplotypes, where **A** is the most parsimonious haplotype network illustrating the inter-relationship between geographical iso-

lates within and between lineages 1 and 2, and **B** illustrates the frequency of haplotypes in total numbers

Fig. 5 Pairwise mismatch distribution curves for COX1 haplotypes within each of the *Tarebia granifera* lineages. Values of expected frequency of pairwise mismatches are shown with a dashed line and the frequency of observed matches are a solid black line. **A** represents lineage 1, **B** represents the invasive sub-clade A from lineage 2 and **C** represents sub-clade B from lineage 2. All graphs indicate high frequencies of pairs of haplotypes with low divergence and rapid population expansion

or incomplete lineage sorting commonly seen in recently radiated invertebrate species complexes (Pentinsaari et al. 2017; Pérez-Portela et al. 2013; Zhang et al. 2013). Nonetheless, it is important to note that the Hebert et al. (2003) 3% threshold of distinguishing between species is heuristic and was employed to provide an initial insight into the divergence between lineages and sub-clades in the context of the utility of COX1 DNA barcoding. Conversely, the 4× ratio criterion is developed from population genetic theory and was developed specifically for the delimitation of asexual organisms and has been applied extensively to mitochondrial markers owing to their maternal mode of inheritance (Birky et al. 2010). The criterion had been previously applied to other thiarid snail species within the genus *Aylacostoma* which were recognised as genetically distinct species based on mitochondrial markers (Vogler et al. 2014). Similarly, the 4× ratio analysis has also been applied to other asexual species including rotifers (Birky et al. 2010), crustacea (Cheng and Sha 2017; Copilaş-Ciocianu et al., 2022; Marrone et al. 2013; Martens et al. 2012; Schön et al. 2012) and complete mitochondrial genomes of trematodes (Jones et al. 2020). Thus, the 4× ratio was directly applicable to the mitochondrial COX1 marker of the predominantly clonal Thiaridae. However, although rare, sexual reproduction does occur occasionally in *Tarebia*, and despite the congruence between the speciation delimitation analyses which supports the notion that *T. granifera* lineage 2 could represent two distinct species, it would predominantly need extensive comparative genomic analyses coupled with detailed morphological examination to confirm (Veeravechskij et al. 2018). Therefore, the three MOTUs identified in this study should be considered as putative until further studies have been able to verify them as distinct species.

Cryptic species are common in snail taxa with highly polymorphic morphology, where distinct species with overlapping characteristics are frequently confused as a single species, and alternatively morphological plasticity has led to the erection of multiple species from a single one. Both cases have been seen in a variety of snail families including but not limited to the Physidae (Albrecht et al. 2025; Lawton et al. 2018), the Lymnaeidae (Lawton et al. 2015; Mirfendereski et al. 2021) and the Littorinidae (Xu et al. 2024), and, in each case, this is attributed to a lack of a molecular phylogenetic framework to underpin the systematics of these mollusc families. Thiaridae has an estimated 200 species,



although there are many more taxa named owing to a high level of redundancy with approximately 70% of all named species being inaccurately identified on morphology alone, with every phenotype being described as a distinct species (Glaubrecht 2009; Veeravech-sukij et al. 2018). Subsequently, this will also lead to missing species that are cryptic and further highlights the need for comprehensive molecular systematic analyses to allow detailed revision of the thiarids (Glaubrecht 2009; Veeravech-sukij et al. 2018).

Despite these caveats, the concordance between the speciation delimitation analyses and the population genetic approaches suggests three putative species within the *T. granifera* complex, and this would have direct implications for the invasion biology of the group. Only sub-clade A was found to contain invasive populations both from South Africa and globally, supporting the hypotheses that invasive populations outside the native range of *T. granifera* represent a single lineage of the complex, allowing the rejection of the hypotheses that multiple lineages were able to establish in South Africa. This also raises the further hypotheses that invasion success may be lineage specific rather than a characteristic of the *T. granifera* complex as a whole. Lineage-specific invasion histories have been reported in other mollusc species in which mitochondrial phylogeographic studies have repeatedly shown that only a subset of lineages within a species complex participates in invasions and that these lineages can differ immensely in their spread and ecological impact. The invasive parthenogenic thiarid snail *Melanoides tuberculata* provides the closest comparison with *T. granifera* with genetically and morphologically distinct lineages which coexist within a species complex (Facon et al. 2003). Phylogeographic analyses of mitochondrial genes indicated that there were at least six independent introductions of *M. tuberculata* in the New World, with invasive clones originating from across the native phylogeny rather than a single region of origin (Facon et al. 2003). Similarly, eight randomly distributed COX1 haplotypes found in Florida indicated multiple invasion events by diverse representatives of the species complex (Tolley-Jordan et al. 2023). Molecular phylogenetic analyses of African populations of *M. tuberculata* identified independent invasion events of Asian isolates into Lakes Malawi and Tanganyika, the Congo River, Nigeria and Cameroon, with invasive populations overlapping native African forms in morphospace and thereby masking their spread (Van Bocxlaer et al. 2015). Crucially, each of these studies demonstrated that individual clonal lineages from within the *M. truncatula* complex possess distinct invasion dynamics and that introduced clones may have competitively excluded one another rather than merging into a

single invasive population (Van Bocxlaer et al. 2015; Facon et al. 2003; Tolley-Jordan et al. 2023). This phenomenon is not confined to gastropods and has been reported in other molluscs also. The freshwater clam *Corbicula* consists of multiple morphologically identical clonal lineages, denoted into forms A–D, with invasive distributions which are lineage specific. Within *Corbicula*, form A is globally distributed with form B restricted to western North America, form C occurring in European and South American watersheds and the recently confirmed New World form D appears to be confined to the Illinois River (Haponski and Foighil 2019). The distinct lineages of *Corbicula* which can be separated by mitochondrial markers are able to hybridise where their populations overlap (Pfenninger et al. 2002; Renard et al. 2000), and, in Argentina, two mitochondrial lineages exhibit virtually segregated distributions despite similar invasion histories and origins (Hünicken et al. 2024). In both *M. tuberculata* and *Corbicula*, it was the genetic lineage that was crucial for invasion biology rather than the morpho-species, and this study argues this to be the case for the *T. granifera* complex too. However, the authors agree that testing whether sub-clade A of lineage 2 is intrinsically more invasive than the others with the *T. granifera* complex will require extensive experimental and genomic approaches that are beyond the scope of the present single-marker dataset.

The South African *T. granifera* shared a single COX1 haplotype across all localities sampled, which was also shared with snail populations from China and Malaysia and appeared to be a sister to a haplotype that was shared between snail populations from the USA, Tahiti and Moorea, illustrating the occurrence of only two invasive haplotypes which exist within the same sub-clade. It is important to note that diversity between haplotypes does exist elsewhere within the sub-lineage, illustrating that although there is diversity within sub-clade A, there is a complete lack of haplotype variation across all sampled South African localities. The data presented here would suggest that the South African population of *T. granifera* is derived from a single population from the mainland of Southeast Asia (Veeravech-sukij et al. 2018). Within invasive molluscs, the presence of low genetic diversity or a single mitochondrial haplotype across an entire non-native range is indicative of either a major bottleneck event or that the founding population was small because of a single invasion, and although bottlenecks are expected to reduce establishment probability, some of the most successful invaders have exhibited dramatically reduced genetic diversity (Therriault et al. 2005). As *T. granifera* is parthenogenetic, only a single individual would have been required to establish a new population in a naive

environment. The complete lack of genetic diversity in the South African *T. granifera* population shares parallels with the invasion of the western USA by a single clonal lineage of the New Zealand mud snail *Potamphyrgus antipodarum* where limited introductions were followed by rapid expansion events (Dybdahl and Drown 2011). Similarly, single haplotype invasions have been documented in other aquatic molluscs such as the European populations of the freshwater mussel *Sinanodonta woodiana* (Konečný et al., 2018) and the Hawaiian populations of the apple snail *Pomacea canaliculata* (Yang et al., 2018). Both species were shown to have high levels of genetic diversity within their native ranges and with only single haplotypes found within the non-native environments, this was interpreted as evidence of a single colonisation event as the source of the entire population which led to the rapid expansion of these species across the non-native ecosystems (Konečný et al. 2018; Yang et al. 2018). The data presented in this current study would therefore suggest a single invasion event into South Africa by *T. granifera* which subsequently led to the rapid expansion of the snail into new environments across the country. However, owing to the parthenogenic nature of the *T. granifera* and the mechanisms by which it has been introduced into the new locations, there is a caveat that applies and needs to be considered. The aquarium trade is considered the most likely mechanism for the introduction of non-native freshwater molluscs, responsible for at least 41% of species introduced into new ecosystems (Preston et al. 2022). Small snails are frequently imported and released as unintended hitchhikers of traded plants and ornamental species (Preston et al. 2022), and aquaculture facilities and aquarium releases are recognised as long-standing conduits for non-native gastropod introductions worldwide (Young et al. 2021). This mechanism of movement can readily distribute single mitochondrial haplotypes across vast distances; for example; the European aquarium populations of the planorbid snail *Gyraulus chinensis* carry haplotypes which are genetically close to wild populations in Italy and Albania, and COX1 sequences from UK aquaria are identical to sequences from Taiwan (Saito et al. 2023). Therefore, the authors suggest that it could be possible that *T. granifera* individuals from the same source population, and thus carrying the same mitochondrial haplotype, could have entered South Africa through multiple invasion events. This could account for the wide distribution of the snail throughout South Africa, as well as the rapid impacts these snails have had on local biodiversity and ecology, replacing most of the molluscan fauna within each habitat (Gerber et al. 2024; Roux et al. 2025). The occurrence of cryptic invasion events by aquatic snails is not uncommon, as stated previously, Asian genotypes of *M. tuberculata* replaced native populations of snail in African freshwater ecosystems through multiple independent invasion events (Van Bocxlaer et al. 2015). Such events

went undetected owing to a lack of morphological differences between populations and often owing to a lack of differentiation in the genetic markers used (Van Bocxlaer et al. 2015). Also, the pairwise mismatch analyses and the assessment of raggedness support sub-clade A representing a rapidly expanding population of a highly successful invasive species, conforming to patterns seen in other invasive freshwater snails with a global distribution (Dybdahl and Drown 2011). However, the results of the current study have revealed that single DNA barcoding approaches are not sufficient to truly disentangle the invasion biology of *T. granifera*. To thoroughly address the cryptic invasion concept for these snails', genome-wide markers would be crucial to compare populations from across native and non-native ranges; however, to date, no genome resources are readily available but will become crucial for monitoring the spread of these snails. It is likely that the putative species represented by sub-clade A from lineage 2 of *T. granifera* will continue its spread across the tropics, and it is vital that molecular and genomic tools are developed and employed to not only monitor its spread but also to be able to distinguish between the different cryptic species within *T. granifera*. This will be crucial for assessing the invasive capability of the different cryptic species should any other snails from the different lineages become invasive. This is an essential part of ecological impact assessment and more so for disease surveillance and monitoring of exotic trematode spread, highlighting once again that invasions of *T. granifera* are truly a One Health issue.

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Author contributions S.P.L.: conceptualisation, formal analysis, investigation, methodology, supervision, visualisation, writing—original draft and writing—review and editing. W.M.: conceptualisation, data curation, funding acquisition, investigation, methodology, project administration, resources, supervision, validation, visualisation and writing—review and editing. J.J.P.: conceptualisation, investigation, methodology, resources and writing—review and editing. R.G.: investigation, methodology, resources and writing—review and editing. L. deN.: conceptualisation, methodology, project administration, supervision, validation and writing—review and editing. N.J.S.: conceptualisation, funding acquisition, investigation, methodology, project administration, resources, supervision, validation and writing—review and editing.

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Data availability Published COX1 DNA sequences of *Tarebia granifera* which represented populations from Southeast Asia, Tahiti and the USA were accessed from GenBank (<https://submit.ncbi.nlm.nih.gov/about/genbank/>). The COX1 sequences generated in this study from *T. granifera* were also made available on Genbank post-publication.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Supplementary Table 1

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